

30 DAYS SQL

MICRO COURSE ASSIGNMENT

“QUESTION AND ANSWER”

Q1} TOTAL NUMBER OF PATIENTS.

Write an SQL query to find the total number of patients across all hospitals.

ANSWER: -

```
SELECT SUM (Patient_Count) AS TOTAL_NUMBER_OF_PATIENT
FROM HOSPITALS;
```

Q2} AVERAGE NUMBER OF DOCTORS PER HOSPITAL.

Retrieve the average count of doctors available in each hospital.

ANSWER: -

```
SELECT ROUND(AVG(Doctors_Count), 2) AS
AVERAGE_NUMBER_OF_DOCTOR_IN_HOSPITAL
FROM HOSPITALS;
RESULT: - 26.71 AVE. NUM. OF DOCTORS IN HOSPITALS.
```

Q3} TOP 3 DEPARTMENTS WITH THE HIGHEST NUMBER OF PATIENTS

Find the top 3 hospital departments that have the highest number of patients.

ANSWER: -

```
SELECT Department, Patient_Count
FROM HOSPITALS
ORDER BY Patient_Count DESC
LIMIT 3;
```

Q4) HOSPITAL WITH THE MAXIMUM MEDICAL EXPENSES.

Identify the hospital that recorded the highest medical expenses.

ANSWER: -

```
SELECT Hospital_Name, Locations, Department, Medical_Expenses  
FROM HOSPITALS  
ORDER BY Medical_Expenses DESC  
LIMIT 1;
```

Q5) DAILY AVERAGE MEDICAL EXPENSES.

Calculate the average medical expenses per day for each hospital.

ANSWER: -

```
SELECT Hospital_Name,  
ROUND(AVG(Medical_Expenses) / NULLIF(AVG(Discharge_Date - Admission_Date),  
0), 2) AS DAILY_AVERAGE_OF_MEDICAL_EXPENSES  
FROM HOSPITALS  
GROUP BY Hospital_Name  
ORDER BY Hospital_Name ASC;
```

Q6) LONGEST HOSPITAL STAY.

Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

ANSWER: -

```
SELECT Hospital_Name, Locations, Patient_Count,  
Admission_Date, Discharge_Date,  
(Discharge_Date - Admission_Date) AS LENGTH_OF_STAY  
FROM HOSPITALS  
ORDER BY LENGTH_OF_STAY DESC  
LIMIT 1;
```

Q7} TOTAL PATIENTS TREATED PER CITY.

Count the total number of patients treated in each city.

ANSWER: -

```
SELECT Locations AS CITY,  
SUM(Patient_Count) AS NUMBER_OF_PATIENTS_TREATED  
FROM HOSPITALS  
GROUP BY Locations  
ORDER BY Locations ASC;
```

Q8} AVERAGE LENGTH OF STAY PER DEPARTMENT.

Calculate the average number of days patients spend in each department.

ANSWER: -

```
SELECT Department,  
ROUND(AVG(Discharge_Date - Admission_Date), 2) AS AVERAGE_LENGTH_OF_STAY  
FROM HOSPITALS  
GROUP BY Department  
ORDER BY Department DESC;
```

Q9} IDENTIFY THE DEPARTMENT WITH THE LOWEST NUMBER OF PATIENTS.

Find the department with the least number of patients.

ANSWER: -

```
SELECT Department,  
MIN(Patient_Count) AS Total_Patients  
FROM HOSPITALS  
GROUP BY Department  
ORDER BY Total_Patients ASC  
LIMIT 1;
```

Q10} MONTHLY MEDICAL EXPENSES REPORT

Group the data by month and calculate the total medical expenses for each month.

ANSWER: -

```
SELECT TO_CHAR(Admission_Date, 'MONTH') AS YEAR_2023,  
SUM(Medical_Expensess) AS TOTAL_MEDICAL_EXPENSES  
FROM HOSPITALS  
  
GROUP BY TO_CHAR(Admission_Date, 'MONTH'),  
EXTRACT(MONTH FROM Admission_Date)  
ORDER BY EXTRACT(MONTH FROM Admission_Date) ASC;
```
